- Linear Algebra Exam

"On my honor, I have neither given nor received any unathorized and/or inappropriate assistance for all sessions of this exam. The work done on this exam is totally my own. I understand that by the School code, violation of these principles will lead to a zero grade and is subject to harsh discipline issues."

Murat Mollahaoğlu
150719604

3.) a) $H^2 = (I - 2uu^T)^2 = I^2 - 4uu^T + 4uuu^Tu^T$

since $u^Tu = 1$, $H^2 = I$.

b) since $H = H^T$, $H^TH = 1$ it's prove that H is an orthonormal matrix.

c) Since $Hu = (I - 2uu^T)u = u - 2uu^Tu = u - 2u = -u$, $\lambda = -1$

d) $v^Tu = 0$, $Hu = (I - 2uu^T)u = u - 2uu^Tu = u$, $\lambda = 1$

u has dimension $n-1$, so there is a basis of $(n-1)$ orthonormal eigenvectors with this eigenvalue. Adding in the eigenvector u,

H is diagonalizable.

4.) a) $\frac{dv}{dt} = 10v + 40w - 10v$

$\frac{dw}{dt} = 40w + 10v - 40w$

differencial equation

$\frac{d}{dt}\begin{bmatrix} v \\ w \end{bmatrix} = \begin{bmatrix} 10 & 1 \\ 40 & 1 \end{bmatrix}\begin{bmatrix} v \\ w \end{bmatrix}$

state transition matrix

4) a)

$$\begin{bmatrix} 2 & -2 \\ -1 & 1 \end{bmatrix} \begin{bmatrix} v \\ w \end{bmatrix} \qquad \begin{bmatrix} 2-\pi & -2 \\ -1 & 1-\pi \end{bmatrix}$$

$$= (2-\pi)(1-\pi) - (2)$$

$$= 2 - 2\pi - \pi + \pi^2 - 2$$

$$\pi^2 - 3\pi = 0$$

$$\pi(\pi - 3) = 0$$

$\Rightarrow$ 0 3 $\rightarrow$ eigen values

$\pi_1 = 0 \Rightarrow \begin{bmatrix} 2 & -2 \\ -1 & 1 \end{bmatrix} \times x_1 = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$ $\quad x_1 = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$

$\pi_2 = 3 \Rightarrow \begin{bmatrix} -1 & -2 \\ -1 & -2 \end{bmatrix} \times x_2 = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$ $\quad x_2 = \begin{bmatrix} 2 \\ -1 \end{bmatrix}$

$\rightarrow$ eigen vectors

b) $= U(t) = s_1 \cdot e^{\pi_1 \cdot t} \cdot c_1 + s_2 e^{\pi_2 \cdot t} \cdot c_2 = c_1 + 2c_2 = \begin{bmatrix} 1 \\ 0 \end{bmatrix}$

$c_1 + 2c_2 = 1$

$c_1 - 2c_2 = 0$

$c_1 = 2c_2$

$c_1 + c_1 = 1$

$2c_1 = 1$

$c_1 = 1/2$

$U(t) = 1 \cdot e^{0 \cdot t} \cdot \frac{1}{2} + \begin{bmatrix} 1 \\ -1 \end{bmatrix} e^{2 \cdot t} \cdot 1$ $\quad c_1 = 1/2 \quad c_2 = 1$

$U(1) = 1 e^0 \frac{1}{2} + 2 e^8 \cdot 1 = U(1) = 1 \cdot \frac{1}{2} + 2 \cdot e^8$

C = steady state

$$v(t) = \underbrace{\begin{vmatrix} -1 \\ 1 \end{vmatrix} e^{0 \cdot t} \cdot \frac{1}{2}}_{\lambda_1 = 0 \text{ oldugu için sabit sayıdır}} + \underbrace{\begin{bmatrix} 2 \\ -1 \end{bmatrix} e^{(2) \cdot t} \cdot 1}_{\lambda_2 = 0 \text{ olduğu için } e^{\lambda t} = 0}$$

$$t = \infty = \begin{bmatrix} -1 \\ 1 \end{bmatrix} e^{0 \cdot t} \cdot c_1$$